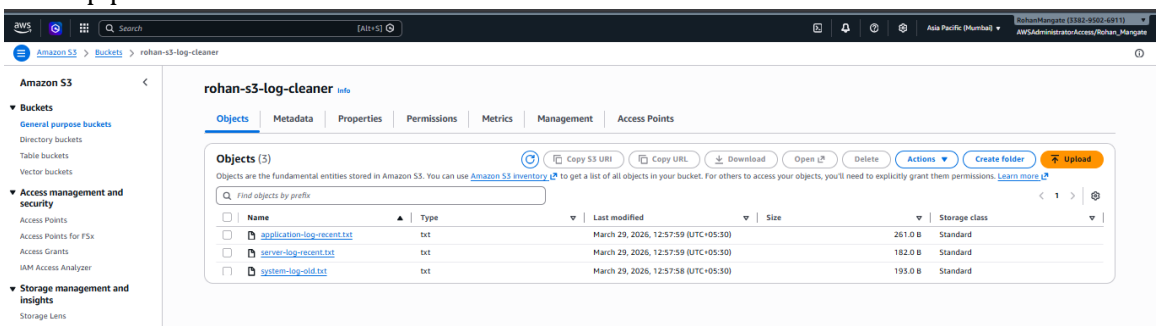
Assignment 15: Implement a Log Cleaner for S3

Objective

The objective of this assignment is to implement an automated log cleanup mechanism using AWS Lambda, Amazon S3, and Amazon EventBridge. The Lambda function periodically scans a specified S3 bucket and deletes log files older than 90 days, thereby optimizing storage usage and cost.

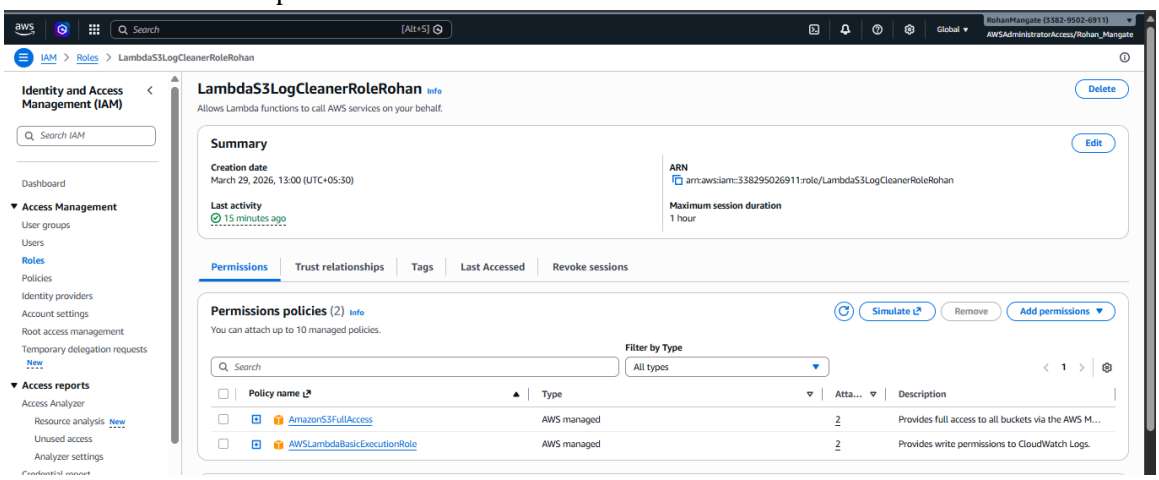
Step 1: S3 Bucket Creation and Log Upload

An S3 bucket was created to store application and system log files. Dummy log files were uploaded to simulate real-world logging data. These files serve as test inputs for the log cleanup process.



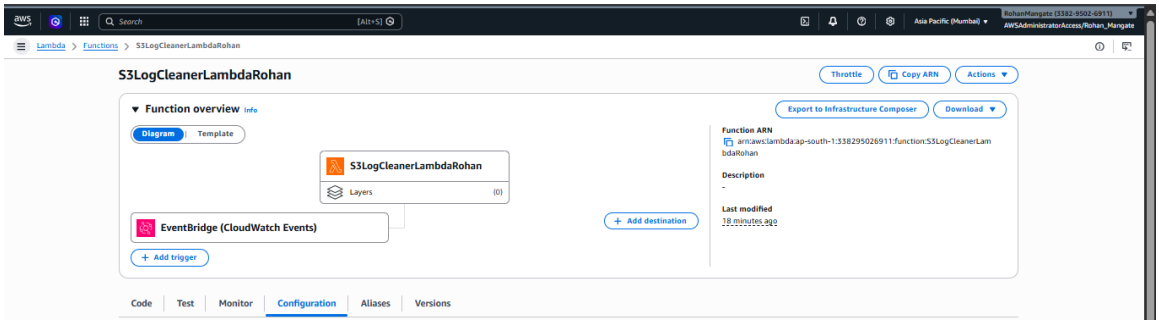
Step 2: IAM Role Creation for Lambda

An IAM role was created to grant the Lambda function permissions to access S3 and write logs to CloudWatch. The role was attached with the AWSLambdaBasicExecutionRole and AmazonS3FullAccess policies.



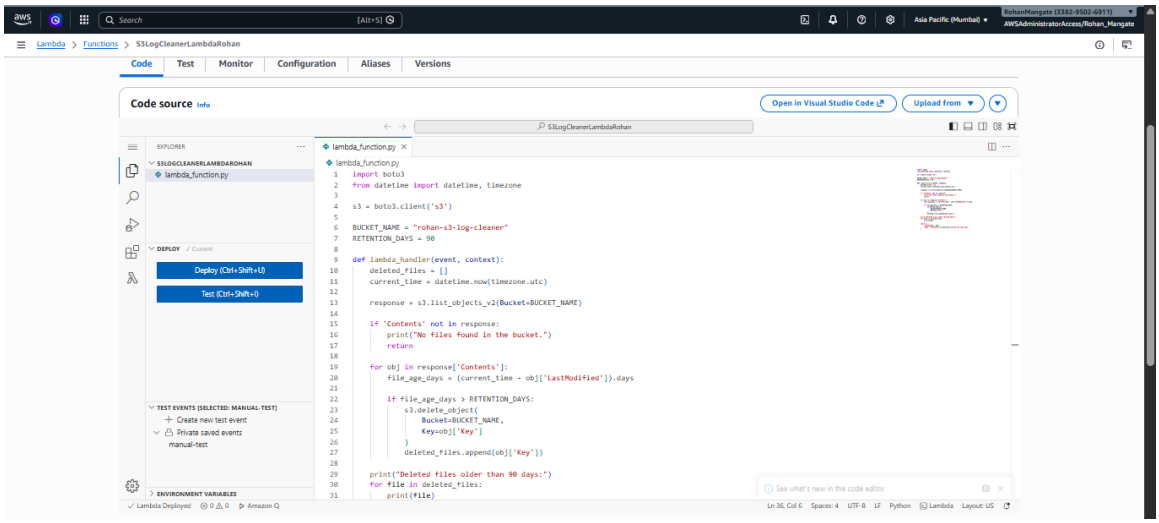
Step 3: Lambda Function Creation

A Lambda function was created using the Python runtime and assigned the previously created IAM role. This function contains the logic to identify and delete log files older than 90 days from the S3 bucket.



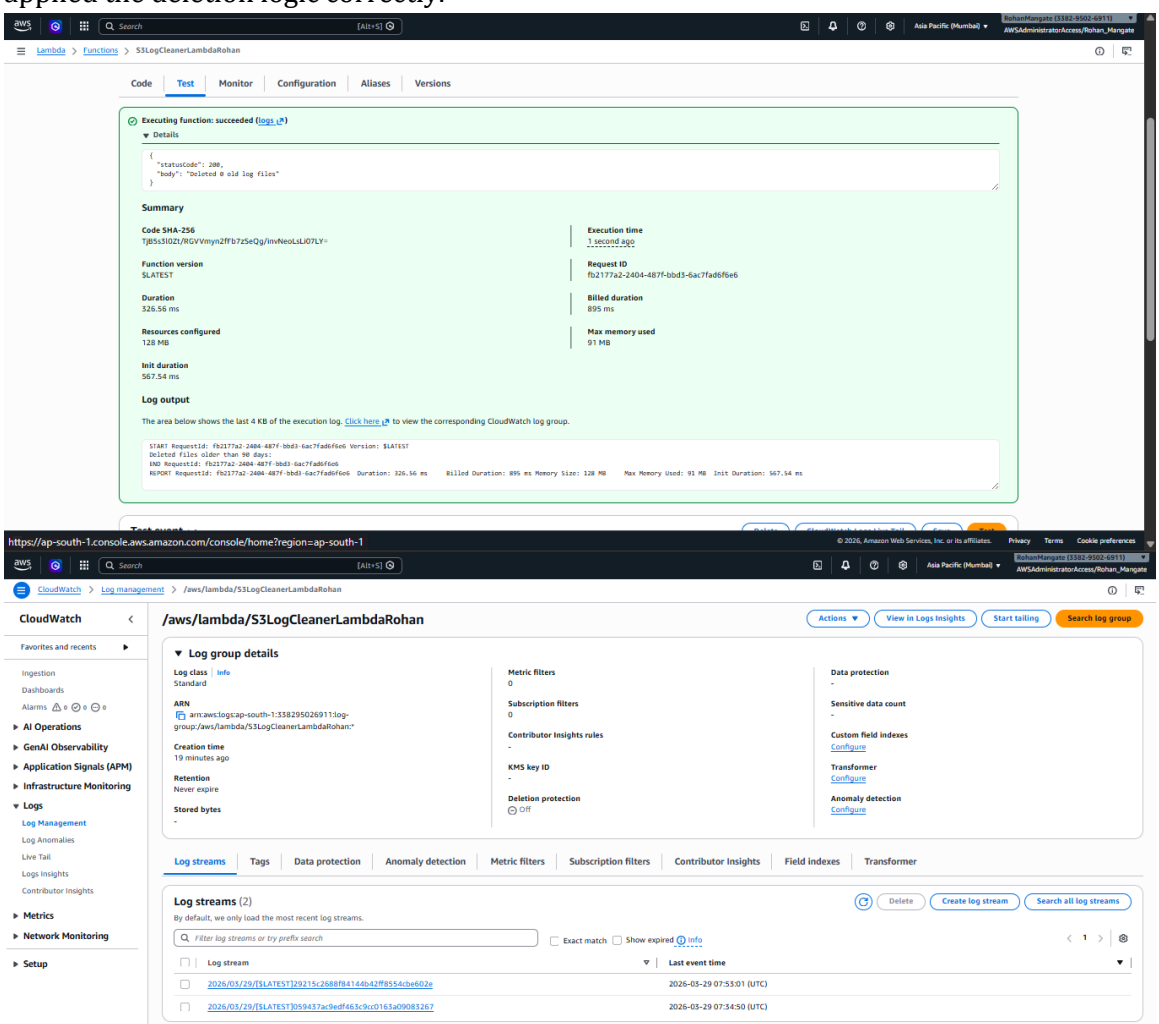
Step 4: Lambda Function Code Implementation

The Lambda function was implemented using Boto3. The code lists all objects in the S3 bucket, calculates the age of each file based on the LastModified timestamp, and deletes files exceeding the defined retention period of 90 days.



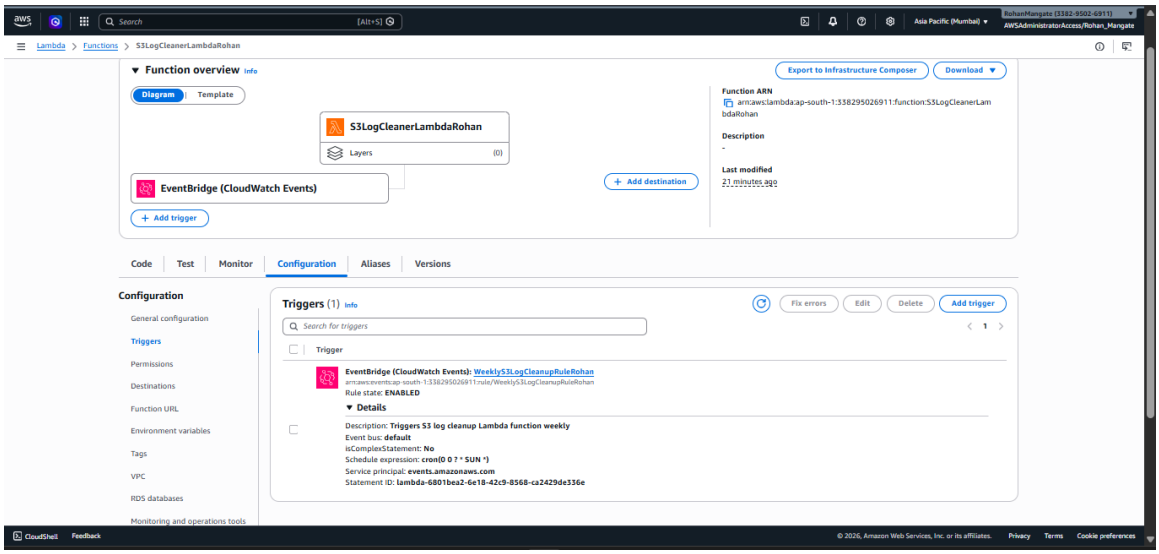
Step 5: Lambda Function Testing and Validation

The Lambda function was manually invoked using a test event. Execution logs were reviewed in Amazon CloudWatch to verify that the function executed successfully and applied the deletion logic correctly.



Step 6: EventBridge Scheduling

An Amazon EventBridge rule was created and attached to the Lambda function to trigger execution on a weekly schedule. The rule is enabled by default, ensuring automatic and periodic log cleanup.



Conclusion

This assignment successfully demonstrates the use of AWS Lambda and EventBridge to automate S3 log cleanup. The solution ensures efficient storage management by deleting outdated log files without manual intervention.